

Package Peak Temperature (°C)

- 3D (all-DRAM): DRAM tier T
- XBM: DRAM tier T
- small box = peak GPU T

thermal headroom

3D thermal penalty

GPU frequency to stay thermally viable

Config	GPU freq	Through-put	Thr. density
3D all-DRAM	0.5×	72% (87%)	122% (146%)
XBM	0.8×	89% (115%)	150% (193%)

first = ref HBM bandwidth · () = expected with 4× bandwidth (3D)

(paper ref.)

Note: identical thermal boundaries for all cases.

STCO approach

